```
#include <GL/glut.h>
#include <cmath>

void init() {
   glClearColor(0.0, 0.0, 0.0, 1);
   glMatrixMode(GL_PROJECTION);
   gluOrtho2D(0, 500, 0, 500);
}

void setPixel(int x, int y) {
   glBegin(GL_POINTS);
   glVertex2i(x, y);
   glEnd();
   glFlush();
}

void bresenham(int x1, int y1, int x2, int y2) {
   int dx = abs(x2 - x1);
   int dy = abs(y2 - y1);
   int x = x1, y = y1;
   int sx = (x2 >= x1) ? 1 : -1;
   int sy = (y2 >= y1) ? 1 : -1;
   int p;

   if (dx > dy) {
      p = 2 * dy - dx;
      for (int i = 0; i <= dx; i++) {
         setPixel(x, y);
         x += sx;
         if (p < 0)
            p += 2 * dy;
         else {
            y += sy;
            p += 2 * (dy - dx);
         }
      }
   } else {
      p = 2 * dx - dy;
      for (int i = 0; i <= dy; i++) {
         setPixel(x, y);
         y += sy;
         if (p < 0)
            p += 2 * dx;
         else {
            x += sx;
            p += 2 * (dx - dy);
         }
      }
   }
```

```
}

void display() {
   glClear(GL_COLOR_BUFFER_BIT);
   bresenham(50, 50, 450, 400);
   glFlush();
}

int main(int argc, char** argv) {
   glutInit(&argc, argv);
   glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
   glutInitWindowSize(500, 500);
   glutCreateWindow("Bresenham Line Drawing");
   init();
   glutDisplayFunc(display);
   glutMainLoop();
   return 0;
}
```